

ELECTRICAL SAFETY

Advisory
(Policy)

Elect is a good servant but an extremely bad master.

- Latter aspect usually experienced when carelessness and ignorance become part of the work at hand.

- The third item besides carelessness & ignorance is the misuse of electricity. These three usually result in injury or death. Therefore adequate knowledge about electricity is ~~not~~ essential not only to the electrician but to the householder & any other person responsible for its use and maintenance.

Danger aspect of electricity.

... Elect is quite dangerous cos it cannot be seen or realised to be present (until it perform something). Its effects are sometime visible but no-one can tell by visual look if a wire is 'Live' or not until it is touched and by then it would be too late.

Duties of an electrician regarding safety.

As an elect. it is significant to ensure that every elect. inst. is safe to use whether by another experienced electrician or a householder or to livestock.

IEE Regs ^{states} recommends that 'Good Workmanship' and ~~the~~ the use of proper materials are essential for compliance with electrical requirements for safety (safe electrical inst.).

Sources & nature of an electric shock

Conditions involved:

- For an electric shock to be perceived, it is necessary for the human body to be in contact with two objects of unequal potentials in such a way that the body forms part of an electrical ^{circuit} in which current flows. The magnitude of such current will then determine the seriousness of the shock (accident).

Of course ~~the~~ an electric shock is dangerous only at ~~at~~ certain minimum values, the degree of danger being dependant on the current and the time for which it flows. Likewise

a low current for a long time is equivalent to a high current for a brief period.

Applied voltage:

~~the~~ the applied voltage is only important in producing the current thro' human body resistance.

Rather interesting, the human body resistance is about 5000 Ω hand to hand or foot to foot. If the body is immersed in a conducting liquid (fluid) the resistance value is lowered to about 200 Ω .

Typical currents and their effects in human bodies.

1-3 milliamperes: causes a threshold of perception.
i.e. a slight tingling sensation is felt.

10-15 mA: the muscles begin to tighten and become difficult to release any object held (i.e. live cdr).

25-30 mA: the muscles are really tight and the person has absolutely no control over them. This is the first dangerous state.

Over 50 mA: At this stage fibrillation of the heart occurs, which is generally lethal if immediate specialised attention is not given.

NB In this respect it realised (visualized) that quite a small current can be fatal particularly if the victim is in contact with a live object has a weak heart.

- Interesting enough the current taken by a 240V, 15W fil. lamp is 62mA.

Common methods of Protection:

The various methods employed for protection of human beings and livestock against the risk of electric shock are includes:

- (a) use of insulation (i.e. screening live parts and keeping live parts out of reach to unauthorised persons).
- (b) Earthing of all ^{exposed} metal parts in an installation.

This is to ensure that such a metal part ~~does~~ is prevented from being electrically charged. Earthing essentially provide a path of low resistance to the ground (earth) for any current which results from fault b/w a live cdr and an earthed metalwork.

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Earthing essentially reduce the current available for passing through human body in the event of an earth leakage current in an installation.

IEE Regs Requirements (recommendations)

The IEE Regs ^(requires) recommends that every domestic inst should be tested ^① at intervals of ^② five yrs or less. In this respect a Completion Certificate and Inspection Certificate must be issued duly filled for every new installation, extension or modification of an existing installation. For latter testings only an inspection Certificate is important as stipulated by the regulations of power undertakings.

Such tests must not only be carried out on the installation itself but ~~also~~ to all the current using appliances used by the consumer.

Below are the basic areas that an inspecting electrician should look for.

- (i) Flexible cables not secure at plugs.
- (ii) Frayed cables.
- (iii) Cables without mech. protection.
- (iv) Use of Unearthed metalwork.
- (v) Circuit over-fused.
- (vi) Poor or broken earth connections & especially signs of corrosion.
- (vii) Unguarded elements of radiant fires.
- (viii) Unauthorized additions to final sub-circuits resulting in overloaded cat cables.
- (ix) Unprotected or Unearthed socket outlets.
- (x) Appliances with earthing requirements - being supplied from 2 pin B.C adaptors.
- (xi) Bellwire (with extra-low voltage insulation) used to carry mains voltage.
- (xii) Use of portable heating appliances in bathrooms.
- (xiv) Broken connectors, such as plugs.
- (xv) Signs of heating at socket outlet-contacts.